

# Staggered toroidal shell blocks: a constructive ropelength bound and its first block correction

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## Abstract

We give explicit unit-thickness representatives of the full-twist links  $T(M, M)$  with  $\limsup_{M \rightarrow \infty} \text{Rop}(T(M, M))/[M(1)]^{3/4} < 10.614$ . The construction combines shell-specific conservative capacities with blocks of equally populated toroidal helices. Alternating half-step phases allow radial spacing  $\sqrt{3}$  within blocks; block boundaries retain spacing two. An exact separation estimate and a full-twist cabling argument certify the geometry and topology. For a block size  $b$  among  $T$  shells, the normalized constructed length has an expansion  $\alpha[1 + a/b + db/T + O(b^{-2} + (b/T)^2 + T^{-1})]$ . In the regime  $b \rightarrow \infty$ ,  $b/T \rightarrow 0$ , its first correction among  $b = \lfloor c\sqrt{T} \rfloor$  is uniquely minimized at an explicitly defined integral constant  $c_* \approx 0.422785$ . The results concern upper constructions and a specified block grammar; neither global ropelength optimality nor publication priority is asserted.

## 1 Conventions, context and scope

Thickness is tube radius and the length of a link is the sum of its component lengths. For smooth embedded closed curves, thickness is the minimum of the curvature radius and half the doubly critical self-distance, including intercomponent separation for a union [1]. Our curves are smooth; polygonal computations used for controls are not substituted for their reach. The crossing number of  $T(M, M)$  is  $M(M - 1)$ .

Concentric helices, toroidal bending, shell population choices and torus-hole doubling occur in the work of Klotz and Thompson [2, 3]. Toroidal and hexagonal filament packings have older precedents [4, 5, 6]. These ingredients are not claimed as new. The focused contribution considered here is the exact shell-specific capacity, block phase compatibility, and their quantitative limiting and boundary-error analysis. In particular, reciprocal-plus-linear optimization and a square-root shell scale already occur in [2, Eq. (17)–(18)]. The possible distinction is the geometry-specific expansion proved below. The bounded prior-art audit accompanying this draft labels publication novelty *possibly novel*, not established priority.

Klotz’s common-hole optimized coefficient near 11.68 is conditional on its no-overlap approximation [3]. Our comparison preserves that qualification. No universal lower-bound improvement, matching asymptotic lower bound, or best-known record across other torus families is claimed.

## 2 Finite construction and clearance

For  $R > r > 0$  and a phase  $\phi$  put

$$F_{r,\phi}(t) = ((R + r \cos t) \cos(t + \phi), (R + r \cos t) \sin(t + \phi), r \sin t), \quad t \in \mathbb{R}/(2\pi\mathbb{Z}). \quad (1)$$

Write  $h = R - r$  and  $A(r, h) = \pi r h / \sqrt{r^2 + h^2}$ .

**Lemma 1.** *If  $r \geq 2$ ,  $R \geq 2r + 2$ , and  $N = \lfloor A(r, R - r) \rfloor - 1$ , the  $N$  curves with phases  $2\pi j/N$  have pairwise distance at least two. Each has thickness at least one.*

*Proof.* Here  $N \geq 4$  since  $r \geq 2$ ,  $h \geq 4$  and  $A \geq 8\pi/\sqrt{20} > 5$ . For parameter difference  $u = t - s$  and phase difference  $\Delta$ , direct expansion gives

$$|F_{r,\phi}(t) - F_{r,\psi}(s)|^2 = 4r^2 \sin^2(u/2) + 4(R + r \cos t)(R + r \cos s) \sin^2((u + \Delta)/2).$$

Replacing the positive radial product by  $h^2$  and combining cosines bounds this below by  $2(S - \sqrt{S^2 - 4r^2 h^2 \sin^2(\Delta/2)})$ , where  $S = r^2 + h^2$ . This is at least four if  $r^2 h^2 \sin^2(\Delta/2) \geq S - 1$ . The least nonzero phase has  $\sin^2(\Delta/2) \geq \sin^2(\pi/N)$ . For  $N \geq 3$ ,  $\sin(\pi/N) \geq \pi/(N + 1)$ : use  $\sin x \geq x - x^3/6$  and  $\pi^2(N + 1) \leq 6N^2$ . Our choice  $N + 1 \leq A$  consequently gives the stronger lower bound  $S$ .

For one curve  $|F'| \geq h$  and  $|F''| \leq B = R + \sqrt{5}r$ . Since  $r \leq h - 2$ ,  $h \geq 4$  and  $\sqrt{5} < 9/4$ ,  $B/h^2 \leq 17/(4h) - 6/h^2 \leq 11/16$ ; curvature is at most  $11/16$ . A doubly critical chord with shorter cyclic separation  $0 < u \leq \pi$  satisfies  $u \geq 2h/B$  by Taylor's integral remainder applied to its scalar product with  $F'(s)$ . The displayed distance identity with  $\Delta = 0$  therefore gives chord length at least  $2h \sin(h/B) \geq (5/3)h^2/B \geq 80/33 > 2$ . The longitude is one-to-one and cylindrical radius positive, proving embeddedness. The smooth thickness formula completes the proof.  $\square$

**Lemma 2.** *For shell radii  $u \leq v < R/2$  and any phase difference  $\Delta$ , all point pairs have squared distance at least*

$$(u - v)^2 + 4K \sin^2(\Delta/2), \quad K = \frac{uv(R - u)(R - v)}{uv + (R - u)(R - v)}.$$

For  $0 < r_0 \leq u \leq v < R/2$ , consider a block with common population  $N = \lfloor A(r_0, R - r_0) \rfloor - 1$ ,  $r_0$  its first radius and  $N \geq 3$ . Alternating phase offsets  $0, \pi/N$  permit neighboring radial spacing  $\sqrt{3}$ .

*Proof.* The transverse squared term is  $(u - v)^2 + 4uv \sin^2((t - s)/2)$ ; the remaining radial product is at least  $(R - u)(R - v)$ . The same harmonic minimization gives  $2(S - \sqrt{S^2 - 4P \sin^2(\Delta/2)})$ , with  $S = uv + (R - u)(R - v)$ ,  $P = uv(R - u)(R - v)$ . Rationalizing, or using  $\sqrt{1 - z} \leq 1 - z/2$ , bounds this by  $4(P/S) \sin^2(\Delta/2)$  from below.

For  $u \leq v < R/2$ , the reciprocal  $1/K = 1/(uv) + 1/((R - u)(R - v))$  decreases as  $v$  increases; its derivative is negative since  $(R - u)(R - v)^2 > uv^2$ . The diagonal value increases with  $u < R/2$ . Thus  $K \geq K(r_0, r_0) = A(r_0, R - r_0)^2/\pi^2$ . For adjacent shells every phase difference is an odd multiple of  $\pi/N$ . Applying the sine inequality with  $2N$  yields

$$4K \sin^2(\Delta/2) \geq \frac{4(N + 1)^2}{(2N + 1)^2} > 1.$$

Adding radial squared gap three proves the assertion.  $\square$

Choose integers  $T \geq 1$ ,  $1 \leq b \leq T$ , let  $\delta = \sqrt{3}$ ,  $g = 2 - \delta$ , and set

$$\begin{aligned} r_i &= \delta i + g \lceil i/b \rceil, & R &= 2r_T + 2, \\ j(i) &= 1 + b \lfloor (i - 1)/b \rfloor, & N_i &= \lfloor A(r_{j(i)}, R - r_{j(i)}) \rfloor - 1. \end{aligned} \tag{2}$$

On each shell use equally spaced phases with offsets alternating  $0, \pi/N_i$  inside its block. Include the core  $C(t) = (R \cos t, R \sin t, 0)$ . There are  $Q = 1 + \sum_i N_i$  components in this bundle. Duplicate the entire bundle by the proper rigid motion  $P(x, y, z) = (R + x, -z, y)$ , giving  $M = 2Q$ .

**Proposition 3.** *For every stated  $T, b$ , this construction has thickness at least one and link type  $T(2Q, 2Q)$  up to a common mirror.*

*Proof.* The first radius is two. Lemmas 1–2 cover same-shell and adjacent within-block pairs. The common block population is conservative for its other shells because  $A(r, R - r)$  increases for  $r < R/2$ . Nonadjacent shells have radial gap at least  $2\sqrt{3}$ , and block boundaries have gap two; the transverse distance alone suffices in these cases. Each shell has distance  $r_i$  from its core. The two round Hopf cores have minimum distance  $R$ , as is seen from

$$|C(t) - P(C(s))|^2 = R^2[1 + 2(1 - \cos t)(1 + \cos s)] \geq R^2.$$

Every component lies within  $r_T$  of its core, so every cross-bundle pair has distance at least  $R - 2r_T = 2$ . The self-thickness estimate applies since  $R \geq 2r_i + 2$ .

For topology use zero-framed tubular coordinates  $E(u, a + ib) = ((R + a) \cos u, (R + a) \sin u, b)$ . Writing  $u = t + \phi$ , every strand is  $E(u, e^{iu}w)$  for a fixed distinct point  $w = re^{-i\phi}$ , including  $w = 0$  for the core. A path of distinct disk configurations induces an isotopy of the link; move these points to a small circle to obtain the closure of the full twist on  $Q$  strands. The radial/vertical frame is the zero framing: a fixed radial push-off is a planar circle with linking number zero.

To check the doubling, rotate two small disjoint point clusters once. The paths  $e^{iu}(c_k + w_{kj})$  factor into rotation of their centers and internal rotation of each cluster. Independent center and internal angles range over a square of collision-free configurations; its diagonal is homotopic to its two edges. Thus a full twist on  $2Q$  points factors as the internal full twists and the zero-framed mutual Hopf twist. The explicit core arrangement is the standard round Hopf link. Signs agree: a first-bundle helix pierces its core's positively oriented spanning disk once downward, and the second core also pierces that disk once downward. The proper motion preserves the other internal sign. Hence this geometric cabling is the full twist on  $2Q$  strands, up to a common mirror. This is a configuration-space and framing proof, not identification by pairwise linking numbers alone.  $\square$

### 3 Leading coefficient and a uniform first correction

Define smooth functions and constants

$$\begin{aligned} n(x) &= \frac{2\pi x(2-x)}{\sqrt{x^2 + (2-x)^2}}, & e(x) &= \int_0^{2\pi} \sqrt{(4 + 2x \cos t)^2 + 4x^2} dt, \\ q_0 &= \int_0^1 n(x) dx = \pi \left( \frac{3 \operatorname{arsinh}(1)}{\sqrt{2}} - 1 \right), & l_0 &= \int_0^1 n(x) e(x) dx, \\ n_1 &= n(1) = \sqrt{2}\pi, & J &= \int_0^1 n'(x) e(x) dx, \\ \alpha &= \sqrt{\delta/2} \frac{l_0}{\sqrt{2}q_0^{3/2}}, & a &= \frac{g}{2\delta}, \quad d = \frac{3n_1}{4q_0} - \frac{J}{2l_0}. \end{aligned}$$

Let  $L(T, b)$  be the exact total analytic length in Proposition 3, and  $F(T, b) = L(T, b)/[M(T, b)(M(T, b) - 1)]^{3/4}$ .

**Theorem 4.** *As  $b \rightarrow \infty$  and  $b/T \rightarrow 0$  through admissible integers,*

$$F(T, b) = \alpha \left[ 1 + \frac{a}{b} + \frac{db}{T} + O(b^{-2} + (b/T)^2 + T^{-1}) \right], \quad (3)$$

*with uniform implied constants in that regime. For fixed  $c > 0$  and  $b = \lfloor c\sqrt{T} \rfloor$  the coefficient of  $T^{-1/2}$  is  $\beta(c) = \alpha(a/c + dc)$ . It is uniquely minimized at  $c_* = \sqrt{a/d} < 1$ , with  $\beta_* = 2\alpha\sqrt{ad} < \beta(1)$ .*

*Proof.* Put  $\lambda = \delta + g/b$ . The exact difference  $r_i - \lambda i = g(\lceil i/b \rceil - i/b)$  lies in  $[0, g)$ , including the incomplete last block. Therefore, uniformly in  $i, T, b$ ,

$$r_i = \lambda i + O(1), \quad R = 2\lambda T + O(1), \quad N_i = \frac{\lambda T}{2} n(j(i)/T) + O(1), \quad \ell_i = \frac{\lambda T}{2} e(i/T) + O(1).$$

Here  $\ell_i$  is one shell component's length. The last two estimates follow from homogeneity, bounded first derivatives of the capacity and speed integral, and an error of less than two for flooring and subtracting one. For example  $|\partial_r A|, |\partial_h A| \leq \pi$ ,  $|\partial_R \ell| \leq 2\pi$ ,  $|\partial_r \ell| \leq 2\pi\sqrt{2}$ . These estimates remain valid at the innermost shells.

Write  $k_i = i - j(i)$ . For any fixed  $C^1$  function  $h$  on  $[0, 1]$ ,

$$\sum_i k_i h(i/T) = \frac{bT}{2} \int_0^1 h(x) dx + O(T + b^2).$$

Indeed, the complete-block lag average is  $(b-1)/2$ . Replacing  $h$  by a block-endpoint value costs  $O(b^3/T)$  per block and  $O(b^2)$  in total. The last incomplete block also costs  $O(b^2)$ . Replacing  $b-1$  by  $b$  and using the Riemann-sum estimate accounts for  $O(T)$ . Taylor expansion of  $n$  at  $i/T$  consequently gives

$$\begin{aligned} \sum_i n(j(i)/T) &= Tq_0 - \frac{b}{2}n_1 + O(1 + b^2/T), \\ \sum_i n(j(i)/T)e(i/T) &= Tl_0 - \frac{b}{2}J + O(1 + b^2/T). \end{aligned}$$

The second identity uses the lag estimate for  $h = n'e$ , not  $(ne)'$ , because each length is evaluated at the actual shell. Including the core,

$$\begin{aligned} Q &= \frac{\lambda}{2} [q_0 T^2 - (b/2)n_1 T] + O(T + b^2), \\ L/2 &= \frac{\lambda^2}{4} [l_0 T^3 - (b/2)JT^2] + O(T^2 + b^2 T). \end{aligned}$$

For  $b/T \rightarrow 0$  the count is comparable to  $T^2$ . This comparability is not asserted for every  $b \leq T$ ; in particular a single block  $b = T$  has only  $O(T)$  components. In the stated regime, expansion of the quotient gives

$$F = \sqrt{\lambda/2} \frac{l_0}{\sqrt{2}q_0^{3/2}} [1 + db/T + O(T^{-1} + (b/T)^2)].$$

The  $M-1$  factor costs only relative  $O(T^{-2})$ . Finally  $\sqrt{\lambda/2} = \sqrt{\delta/2}[1 + a/b + O(b^{-2})]$ ; the cross term is  $O(T^{-1})$ . This proves (3), and flooring  $c\sqrt{T}$  affects the first correction only at order  $T^{-1}$ .

For positivity,  $n' \geq 0$  and the integral triangle inequality and Jensen's inequality give  $8\pi \leq e(x) \leq 2\pi\sqrt{16 + 6x^2} \leq 2\pi\sqrt{22}$ . Hence

$$d \geq \frac{n_1}{q_0} \frac{6 - \sqrt{22}}{8} > \frac{1}{8},$$

because  $q_0 \leq n_1$  and  $\sqrt{22} < 5$ . Also  $0 < a < 1/10$  follows from  $\sqrt{3} > 5/3$ . Thus  $d > a > 0$ . Strict convexity of  $a/c + dc$  on  $c > 0$  and divergence at both ends give the unique minimizer. Moreover  $\beta(1) - \beta_* = \alpha(\sqrt{d} - \sqrt{a})^2 > 0$ .  $\square$

**Theorem 5.** *In the tube-radius convention,*

$$\limsup_{M \rightarrow \infty} \frac{\text{Rop}(T(M, M))}{[M(M-1)]^{3/4}} \leq \alpha < 10.614.$$

*Proof.* Take  $b = \lfloor \sqrt{T} \rfloor$ . The count and length estimates give  $M_T = \delta q_0 T^2 + O(T^{3/2})$  and convergence of the constructed coefficient to  $\alpha$ . For any desired  $M$ , choose the least  $T$  with  $M_T \geq M$ . Monotonicity is not required: minimality gives  $M_{T-1} < M$  and the estimates give  $M_T - M = O(T^{3/2})$ , hence  $M_T/M \rightarrow 1$ . Delete excess components. Restricting a full twist to its surviving points gives  $T(M, M)$ ; thickness cannot decrease and length cannot increase. This proves the all-integer limsup. The strict decimal bound is certified below.  $\square$

The correction theorem compares constructed sequences whose component count changes with  $T, b$ . It does not compare equal- $M$  configurations, establish the exact finite integer minimizer, optimize arbitrary block schedules, or retain a subleading coefficient after the deletion argument.

## 4 Certificates and computational discovery

For  $f(x, t) = n(x)\sqrt{(4 + 2x \cos t)^2 + 4x^2}$  on  $[0, 1] \times [0, 2\pi]$ , direct differentiation gives  $|f_{xx}| \leq 600$  and  $|f_{tt}| \leq 60$ . For completeness the squared speed  $g$  lies in  $[8, 40]$ ; bounds  $|n| < 5$ ,  $|n'| < 14$ ,  $|n''| < 40$ ,  $|g_x| \leq 32$ ,  $|g_{xx}| \leq 16$ ,  $|g_t|, |g_{tt}| \leq 24$  in the chain rule give respective bounds less than 483 and 54. A tensor midpoint rule with  $N$  radial and  $H$  angular cells has error

$$E \leq \frac{2\pi}{24} \left( \frac{600}{N^2} + 60 \left( \frac{2\pi}{H} \right)^2 \right).$$

Arb ball evaluation at exact rational radial nodes and enclosed multiples of  $\pi$  for angular nodes, enclosing all operations, with  $N = 512$ ,  $H = 1024$  and 192-bit precision gives an outward upper endpoint for  $\alpha$  smaller than  $10.613734 < 10.614$ . The artifact stores exact rational endpoints; rounded diagnostic midpoints are never used as inequalities. A separate 224-bit implementation and analytic derivative audit checked the result.

For  $J$  an independent technique sums interval ranges of  $n'(x)$  times the speed over full parameter rectangles, including their exact area. It needs no derivative-error bound. The accompanying boundary-certificate review records its enclosure checks separately from the analytic expansion. Diagnostic high-precision quadrature gives  $c_* \approx 0.4227848052$  and  $\beta_* \approx 3.8835903551$ , versus  $\beta(1) \approx 5.4138303824$ . These displayed approximations are not outward certificate endpoints.

An exhaustive search over every integer  $1 \leq b \leq T$ , for  $T = 8, 16, 32, 64, 128, 256, 512, 1024$ , selected  $b = 1, 2, 2, 4, 5, 7, 9, 13$ . Neither this sequence nor a target exponent was supplied to the objective. The finite fit alone does not prove a square-root law: the last-five-point fitted exponent was approximately 0.425. The smaller block prefactor motivated the expansion in Theorem 4. The search used 48-node Gauss-Legendre length quadrature, with 96-node winner checks; finite normalized values remain numerical diagnostics unless separately certified. Independent reconstruction through  $T = 65539$  tested incomplete blocks and several scaling laws, without finding a counterexample.

## 5 Reproducibility, interpretation and unresolved questions

The repository records generator formulas, geometry and topology proofs, independent adversarial reviews, exact endpoint data, immutable result manifests, software versions and unsuccessful

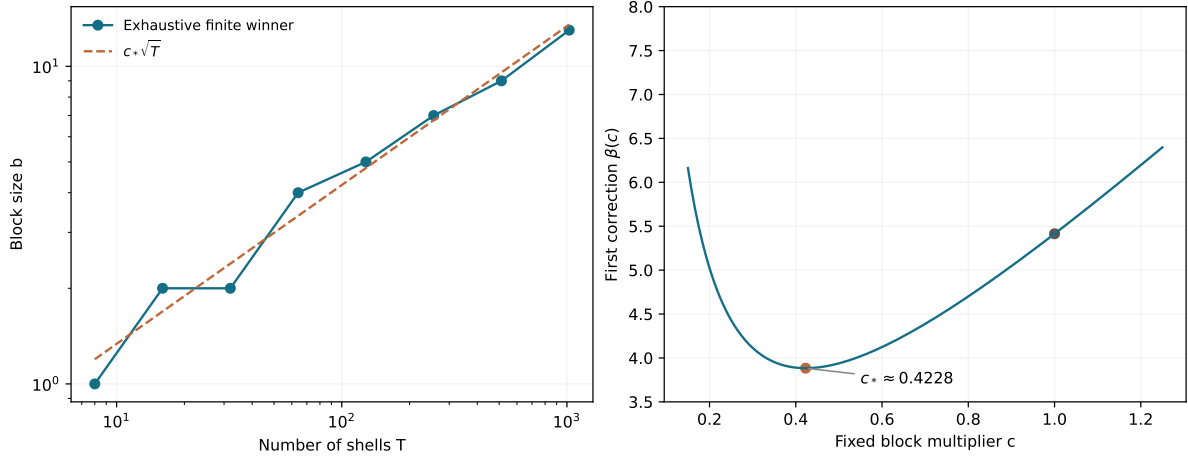


Figure 1: Left: exhaustive finite winners and the asymptotic guide  $c_*\sqrt{T}$ . Right: the analytic first correction  $\beta(c)$ ; its plotted evaluation is numerical. This is an optimization inside a stated construction grammar, not a graph of optimal ropelength.

controls. The local `publication/reproduce.py` checks bundled rational inequalities, reproduces the main integral certificate and creates the figure. Tests, boundary certification and finite block search are explicit additional modes. No external service is needed for the certificate once the pinned Python dependencies are installed.

The physical interpretation is  $M$  separate closed filaments of radius  $r$ : a unit-thickness construction scales to an admissible total contour length  $rL$ . An upper construction gives sufficient length for existence, while a lower theorem gives a necessary length. This does not describe the free-energy minimum of one DNA molecule or a semiflexible polymer. Bending energy, tension, thermal and hydrodynamic effects are additional models.

The outstanding research questions include a finite- $M$  comparison of optimized blocks, variable block schedules, shell-dependent periodic pitch, closure deformations, matching lower bounds and certified criticality. The discovery engine currently exposes a narrower certified shell grammar; its broader numerical extensions must not inherit the present reach theorem without new proofs. The draft is a local publication candidate. Its original contribution and suitability for submission require human scientific judgment; no submission or external contact has taken place.

**AI assistance and review.** The research workflow used delegated Luna workers/testers and Astra adversarial reviewers, with effective model settings logged in the repository. These are independent implementations and adversarial computational reviews within one AI-assisted project, not external peer review or a formal proof-assistant verification. Authorship and publication responsibility have not been assigned.

## References

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